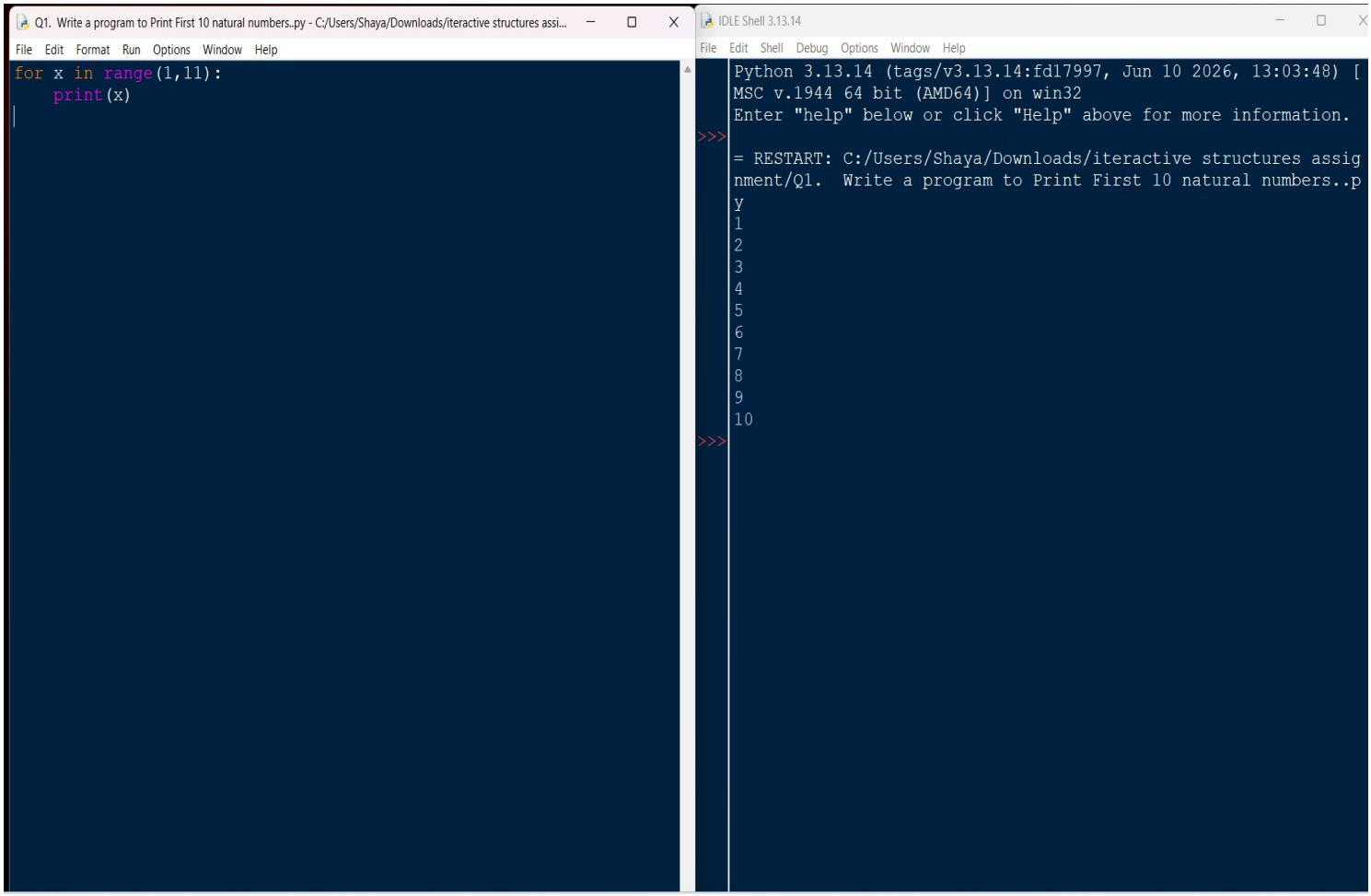
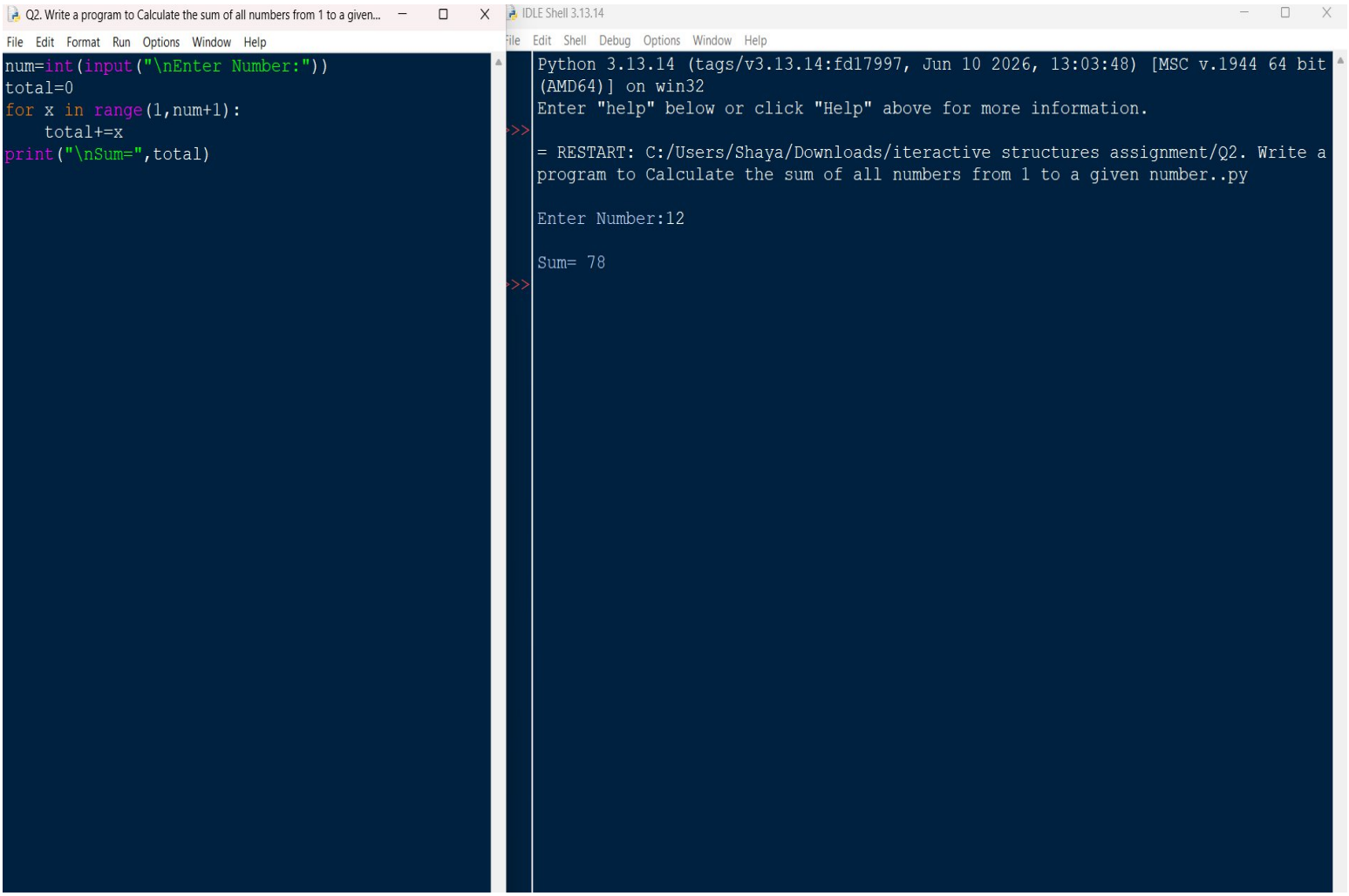
Q1. Write a program to Print First 10 natural numbers.

```
for x in range(1,11):
    print(x)
```



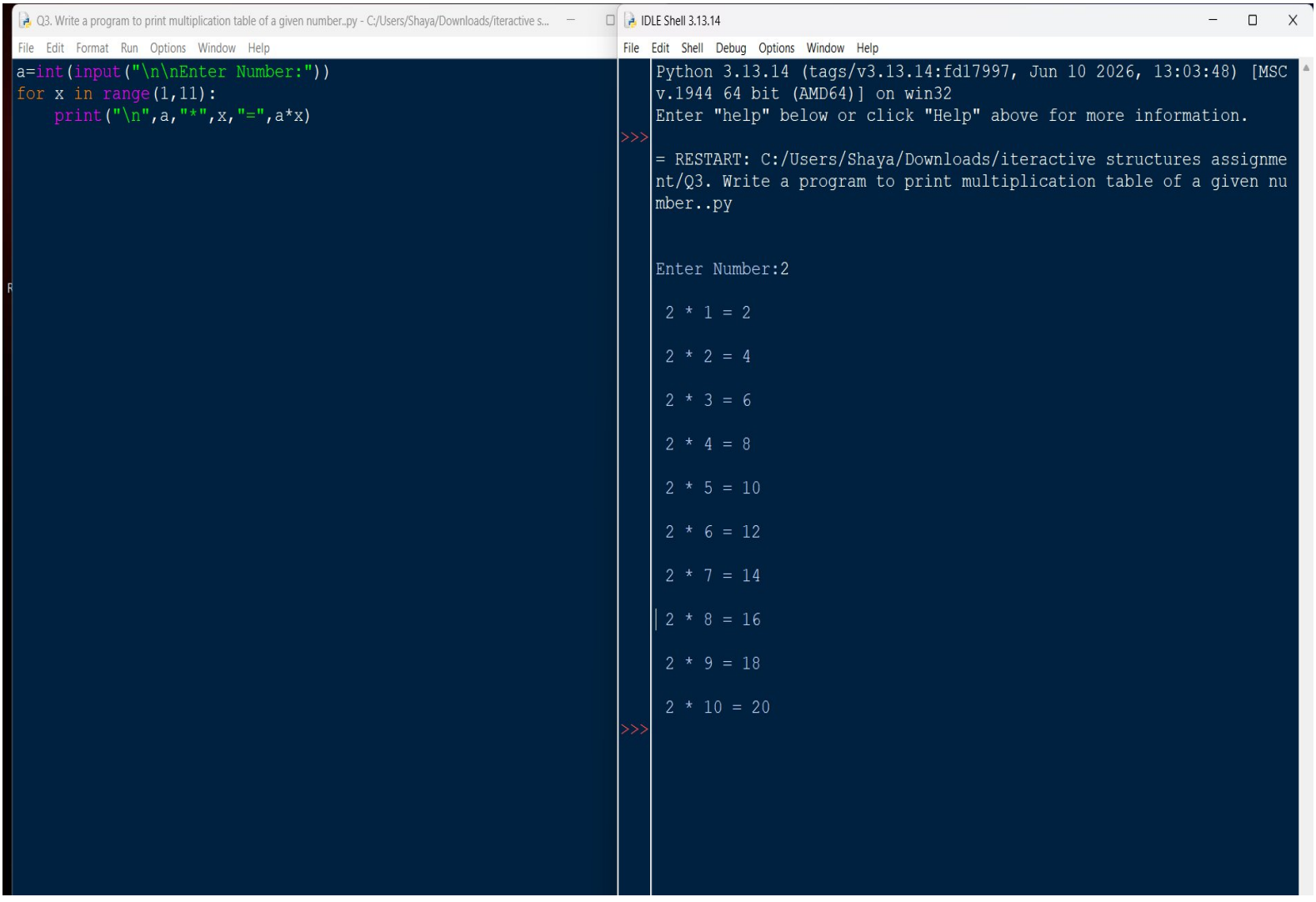
Q2. Write a program to Calculate the sum of all numbers from 1 to a given number.

```
num=int(input("\nEnter Number:"))
total=0
for x in range(1,num+1):
    total+=x
print("\nSum=",total)
```



Q3. Write a program to print multiplication table of a given number.

```
a=int(input("\n\nEnter Number:"))
for x in range(1,11):
    print("\n",a,"*",x,"=",a*x)
```



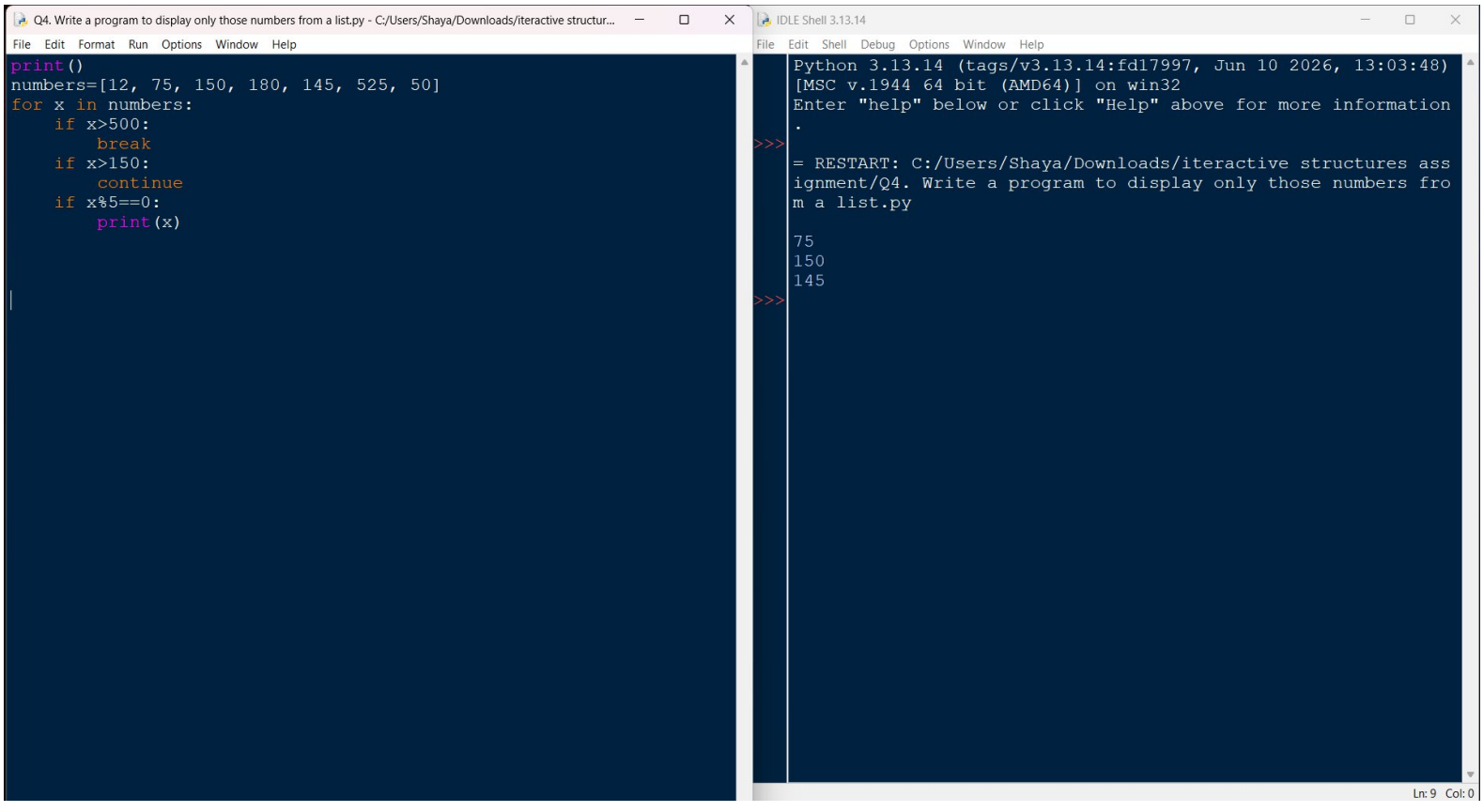
Q4.Write a program to display only those numbers from a list (numbers = [12, 75, 150, 180, 145, 525, 50]) that satisfy the following conditions:

The number must be divisible by five

If the number is greater than 150, then skip it and move to the next number

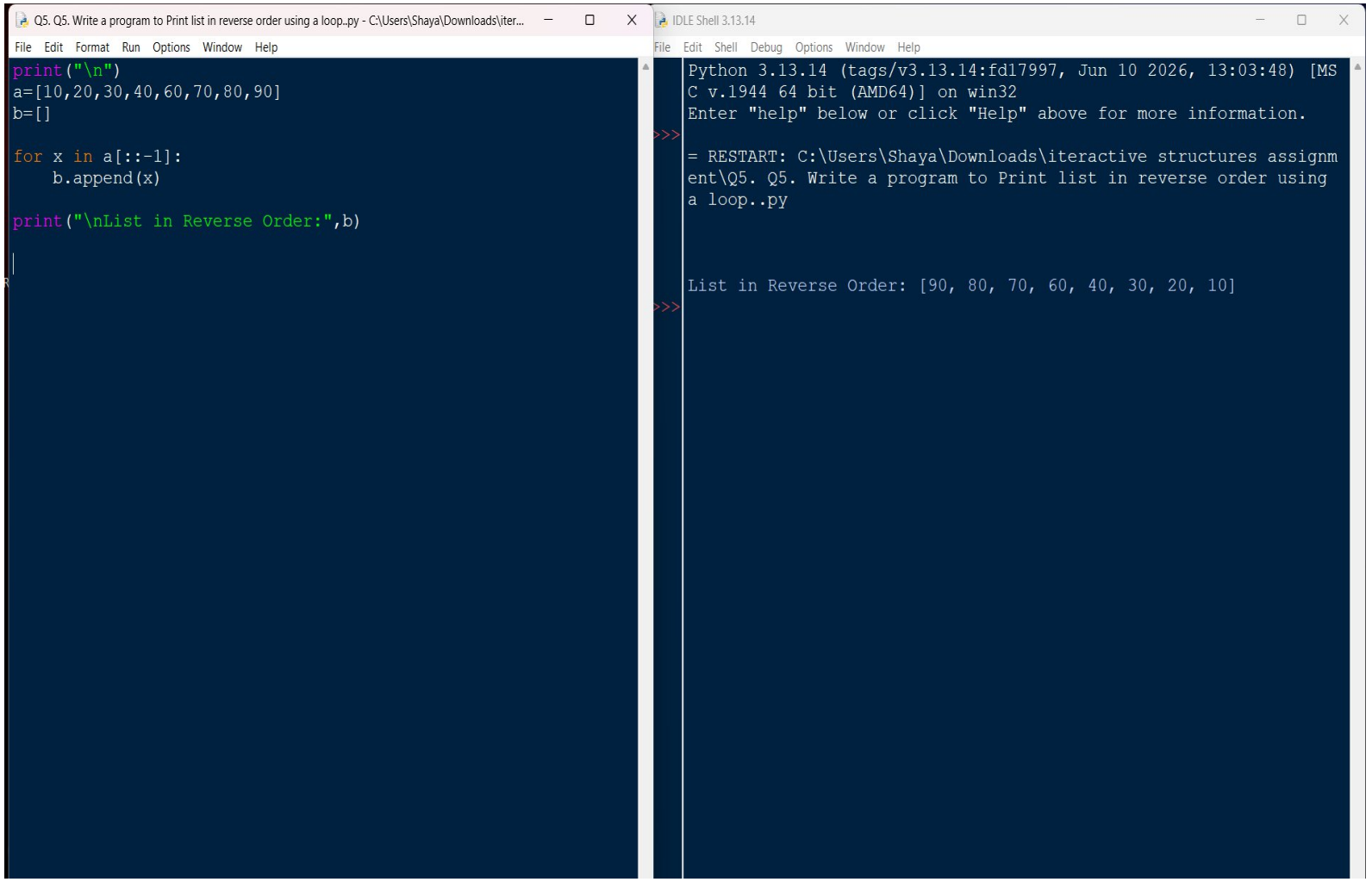
If the number is greater than 500, then stop the loop

```
print()
numbers=[12, 75, 150, 180, 145, 525, 50]
for x in numbers:
    if x>500:
        break
    if x>150:
        continue
    if x%5==0:
        print(x)
```



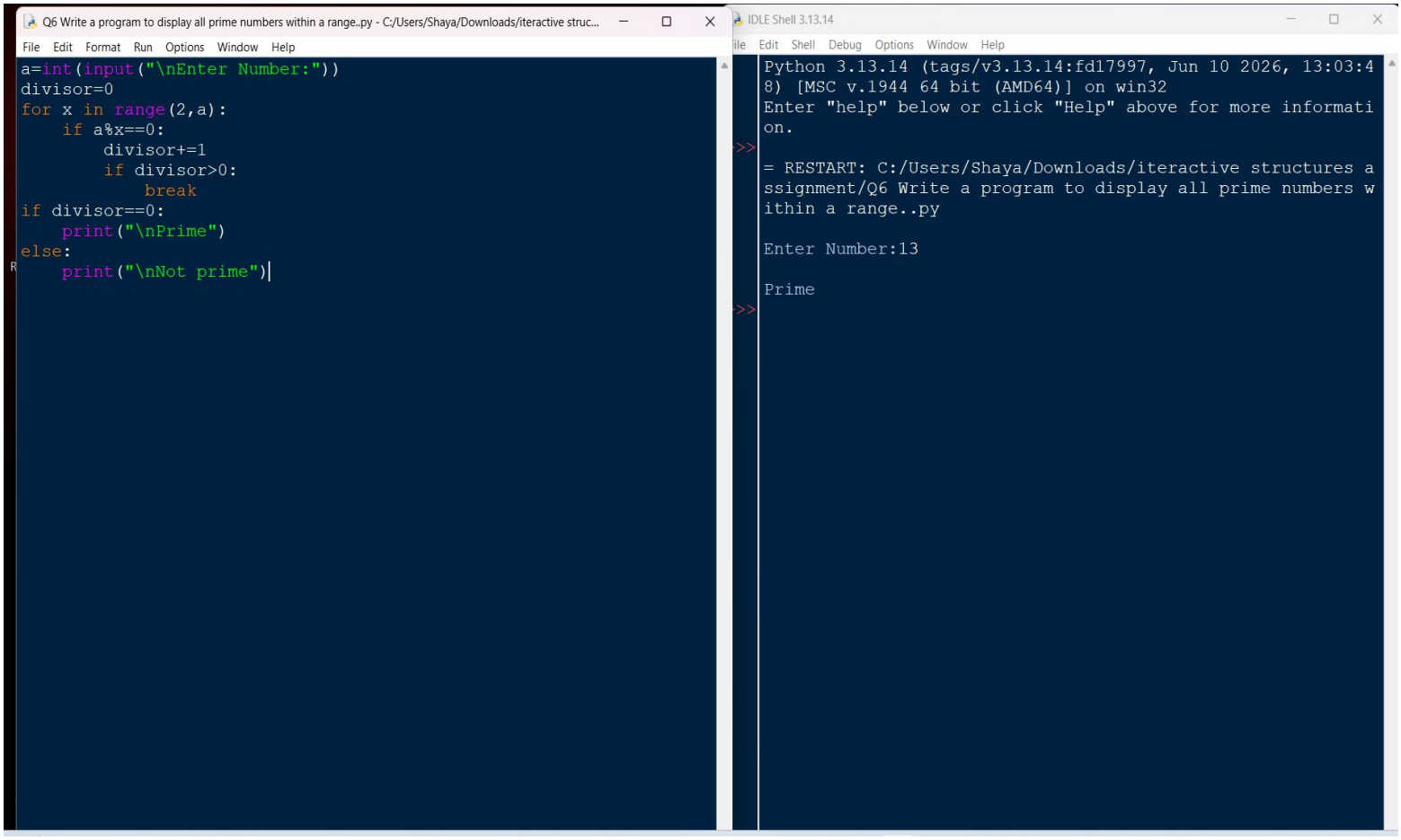
Q5. Write a program to Print list in reverse order using a loop.

```
print("\n")
a=[10,20,30,40,60,70,80,90]
b=[]
for x in a[::-1]:
    b.append(x)
print("\nList in Reverse Order:",b)
```



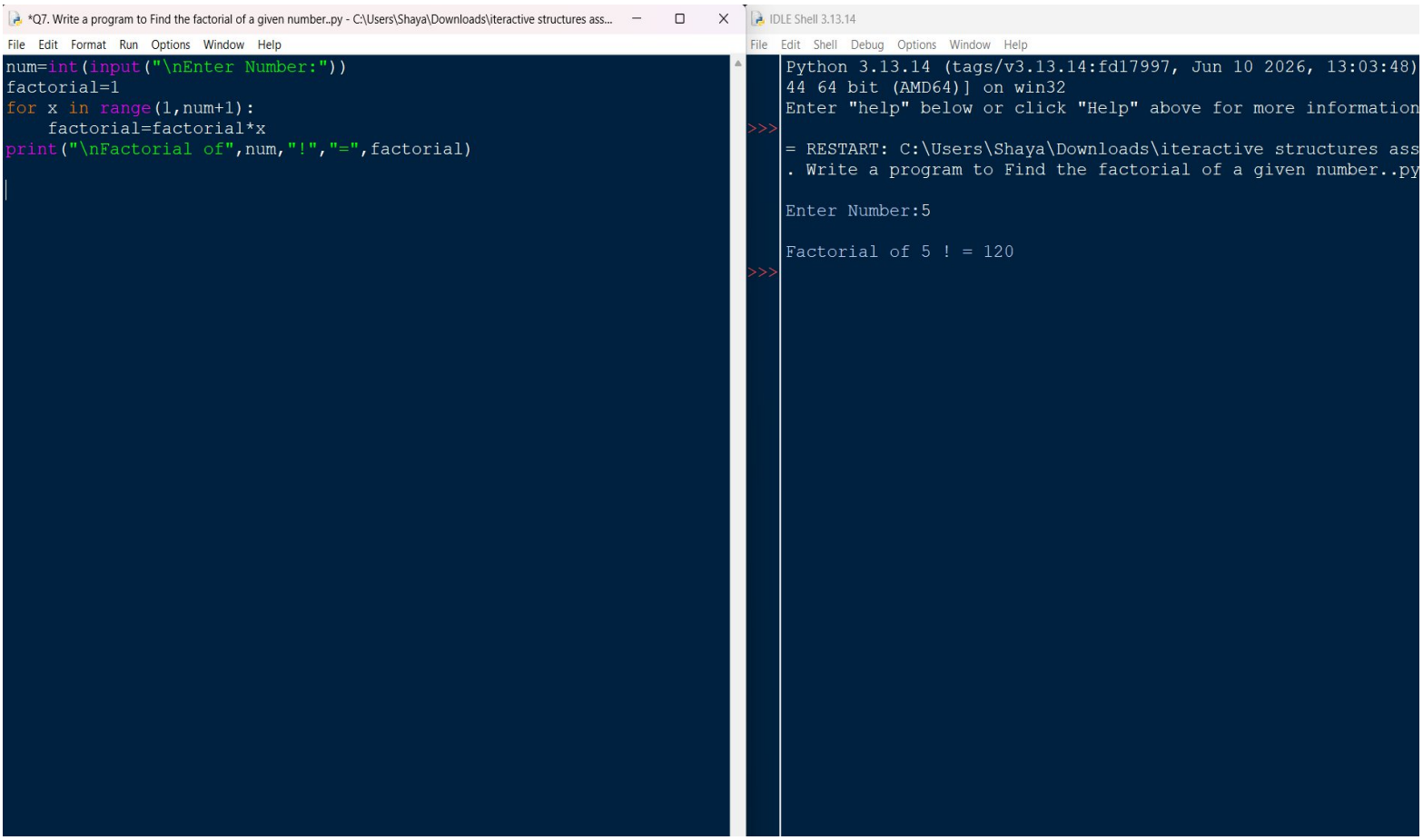
Q6. Write a program to display all prime numbers within a range.

```
a=int(input("\nEnter Number:"))
divisor=0
for x in range(2,a):
    if a%x==0:
        divisor+=1
        if divisor>0:
            break
if divisor==0:
    print("\nPrime")
else:
    print("\nNot prime")
```



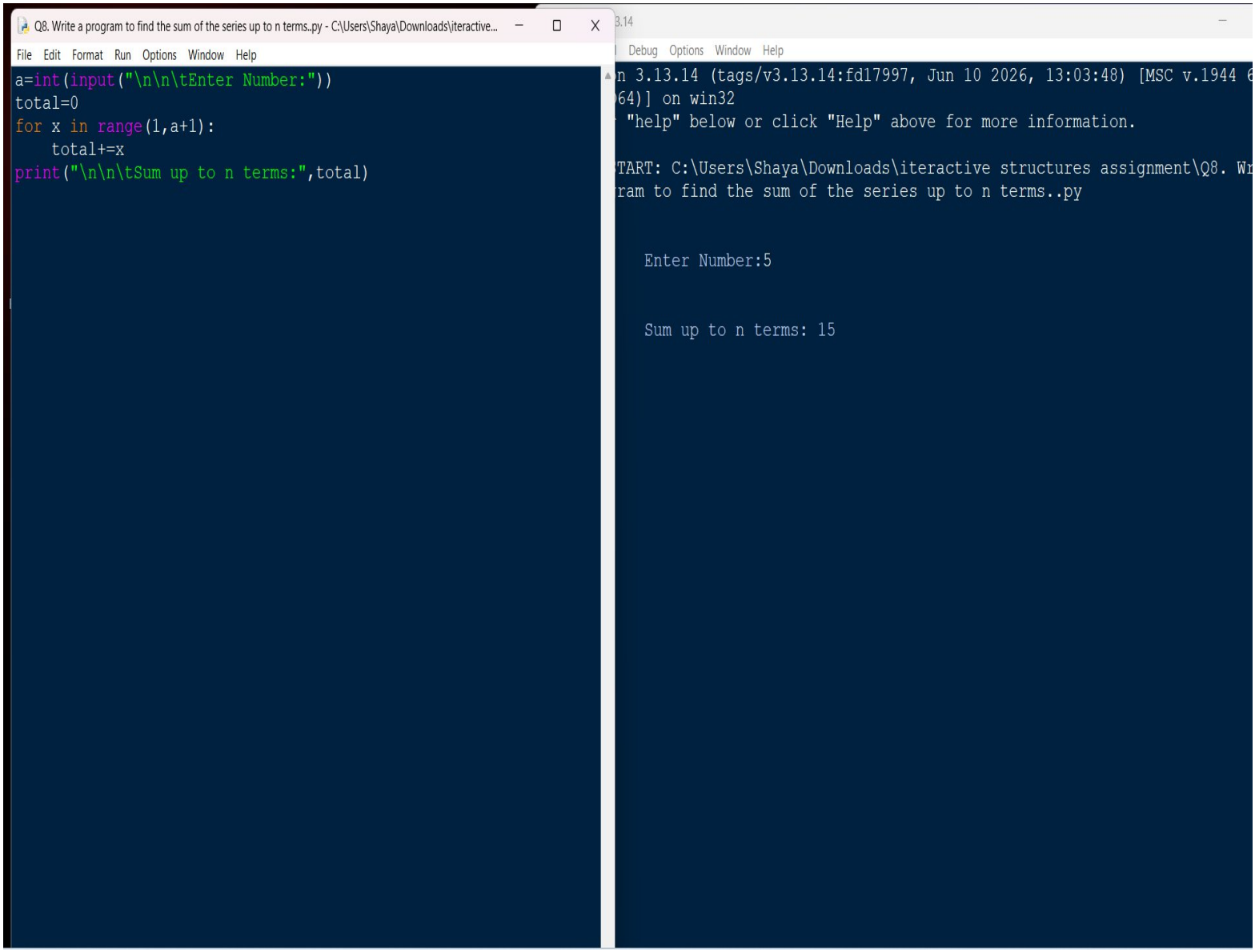
Q7. Write a program to Find the factorial of a given number.

```
num=int(input("\nEnter Number:"))
factorial=1
for x in range(1,num+1):
    factorial=factorial*x
print("\nFactorial of",num,"!","=",factorial)
```



Q8.Write a program to find the sum of the series up to n terms.

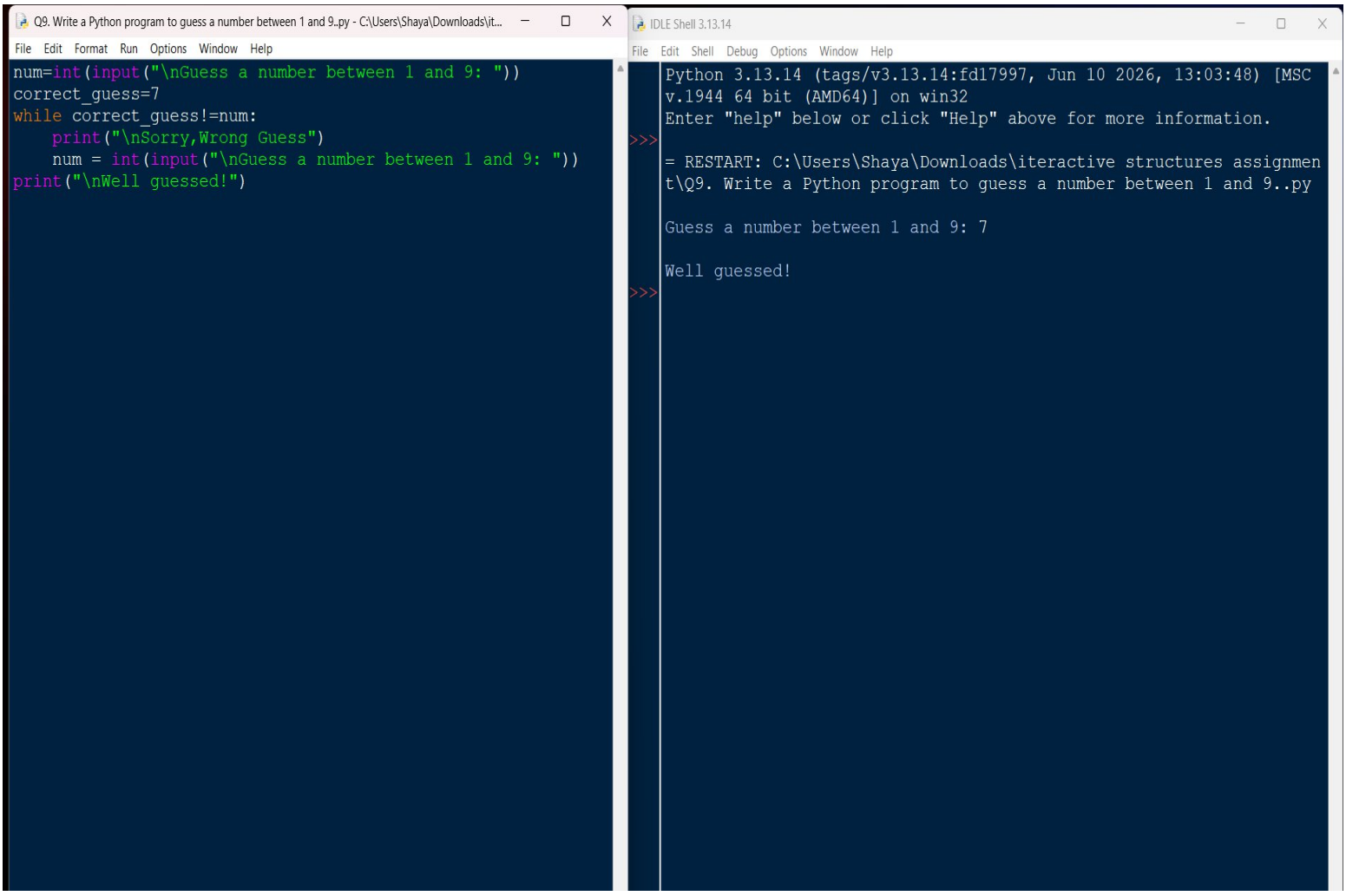
```
a=int(input("\n\n\tEnter Number:"))
total=0
for x in range(1,a+1):
    total+=x
print("\n\n\tSum up to n terms:",total)
```



Q9.Write a Python program to guess a number between 1 and 9.

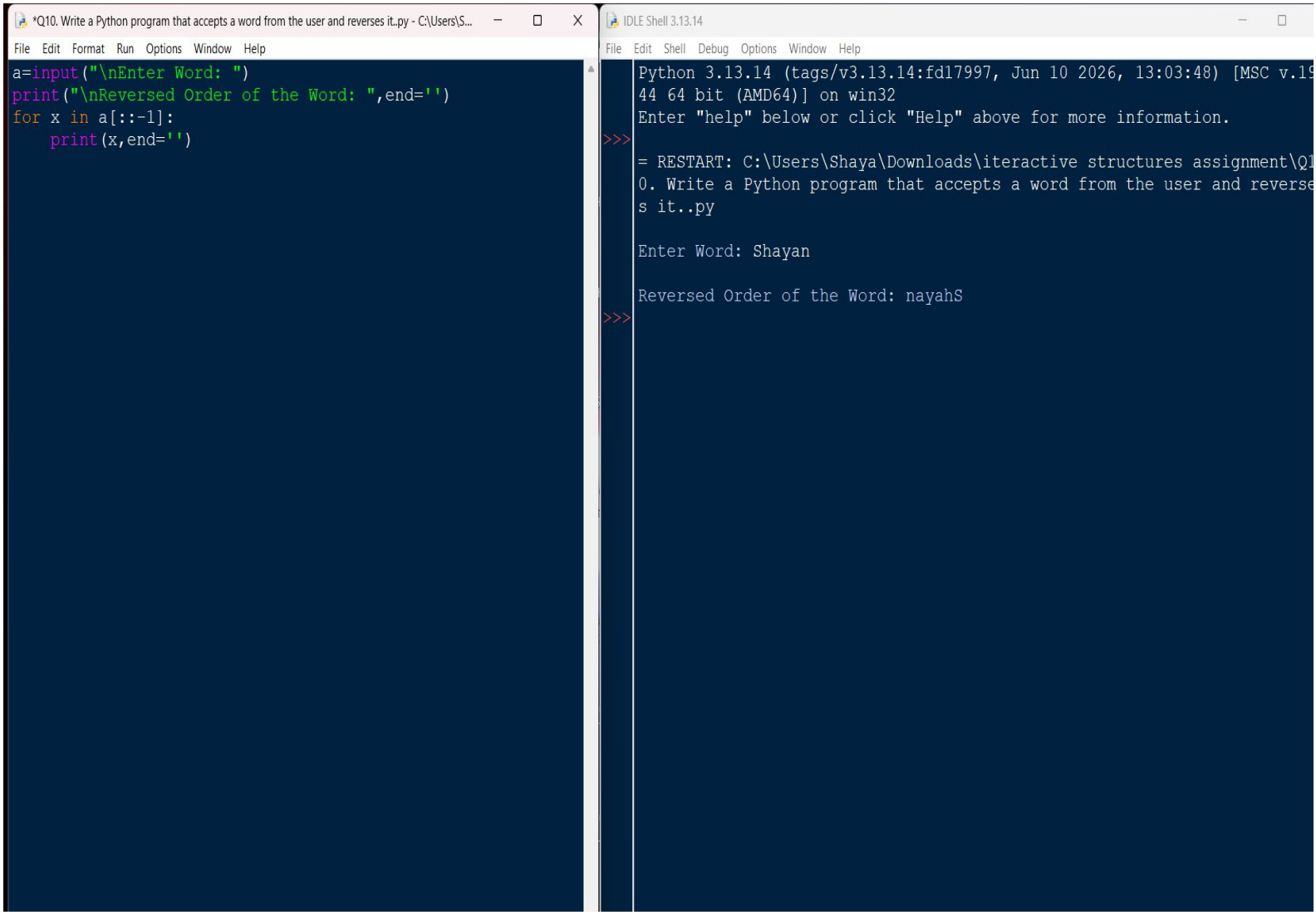
Note : User is prompted to enter a guess. If the user guesses wrong then the prompt appears again until the guess is correct, on successful guess, user will get a "Well guessed!" message, and the program will exit.

```
num=int(input("\nGuess a number between 1 and 9: "))
correct_guess=7
while correct_guess!=num:
    print("\nSorry,Wrong Guess")
    num = int(input("\nGuess a number between 1 and 9: "))
print("\nWell guessed!")
```



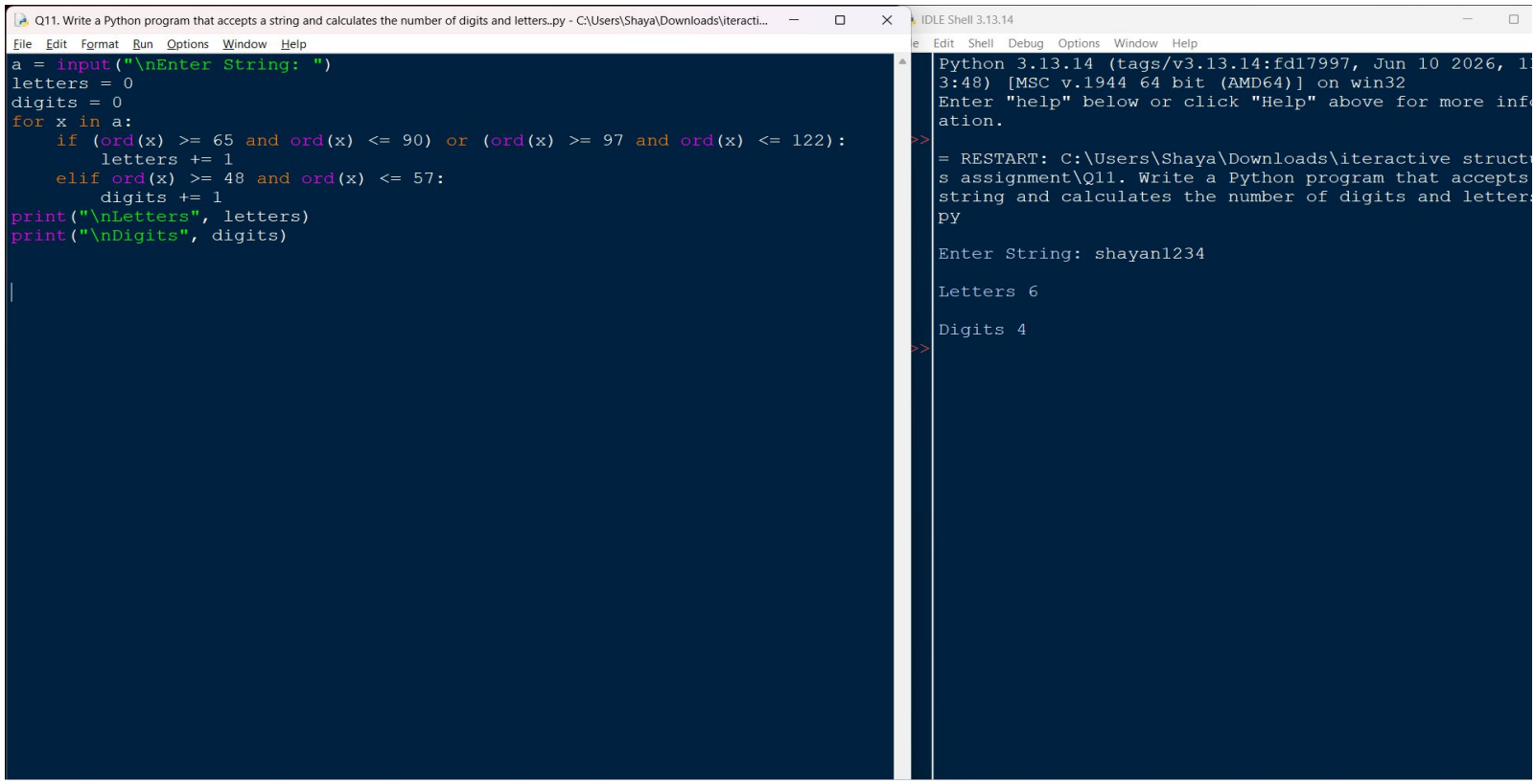
Q10. Write a Python program that accepts a word from the user and reverses it.

```
a=input("\nEnter Word: ")
print("\nReversed Order of the Word: ",end='')
for x in a[::-1]:
    print(x,end='')
```



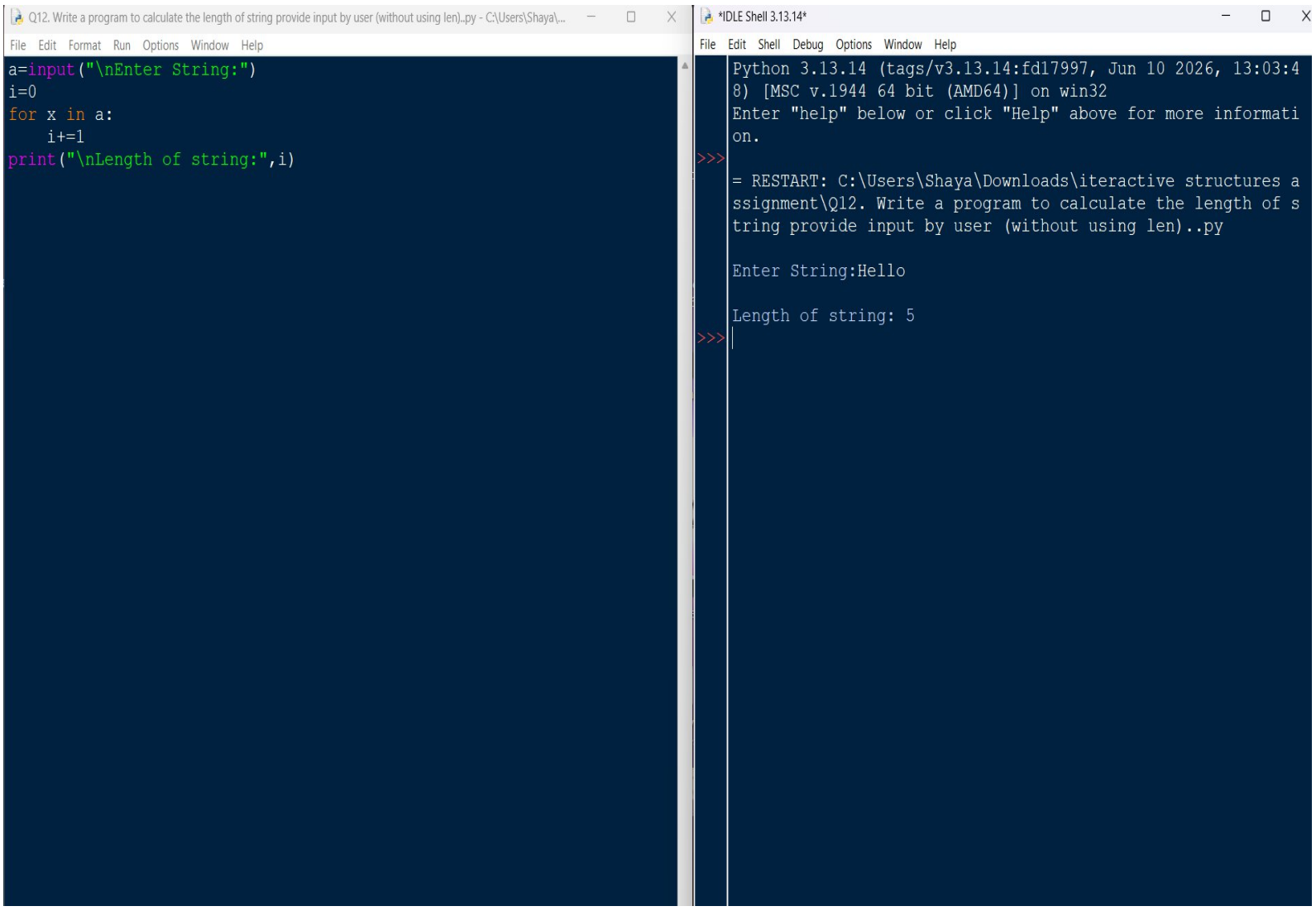
Q11. Write a Python program that accepts a string and calculates the number of digits and letters.
Sample Data : Python 3.2
Expected Output :
Letters 6
Digits 2.

```
a = input("\nEnter String: ")
letters = 0
digits = 0
for x in a:
    if (ord(x) >= 65 and ord(x) <= 90) or (ord(x) >= 97 and ord(x) <= 122):
        letters += 1
    elif ord(x) >= 48 and ord(x) <= 57:
        digits += 1
print("\nLetters", letters)
print("\nDigits", digits)
```



Q12. Write a program to calculate the length of string provide input by user (without using len).

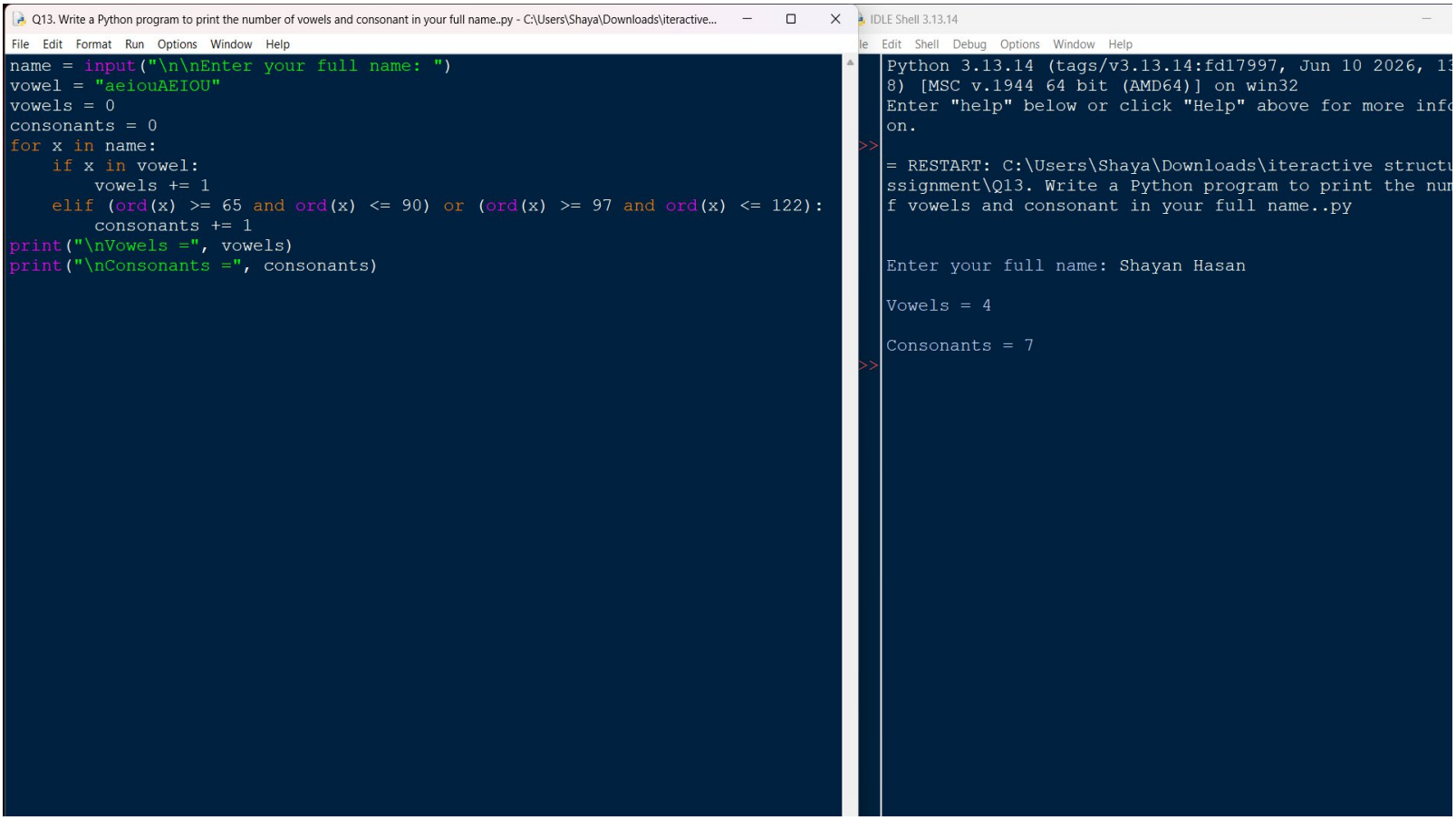
```
a=input("\nEnter String:")
i=0
for x in a:
    i+=1
print("\nLength of string:",i)
```



Q13.Write a Python program to print the number of vowels and consonant in your full name.

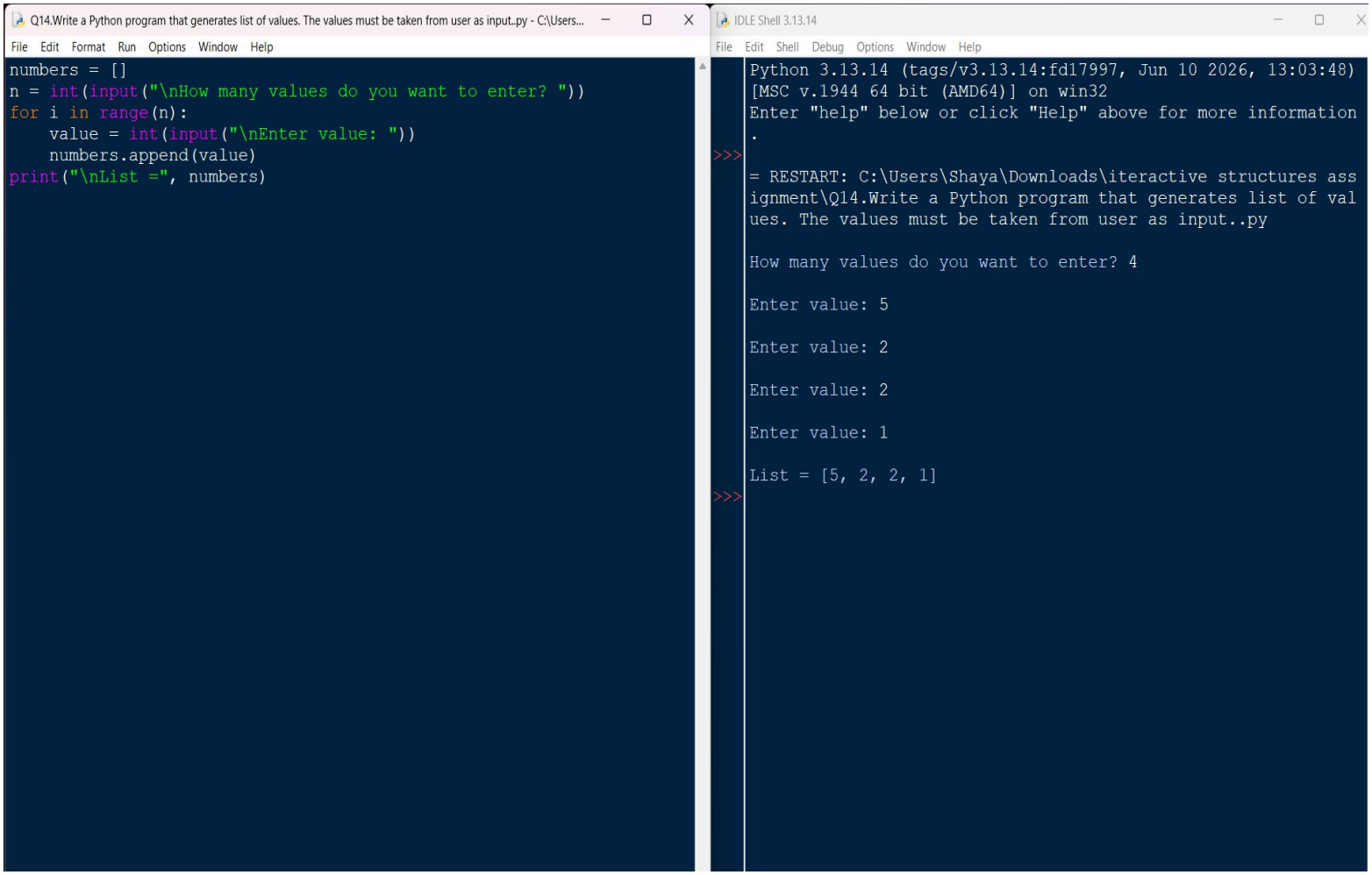
name = input("\n\nEnter your full name: ")

```
name = input("\n\nEnter your full name: ")
vowel = "aeiouAEIOU"
vowels = 0
consonants = 0
for x in name:
    if x in vowel:
        vowels += 1
    elif (ord(x) >= 65 and ord(x) <= 90) or (ord(x) >= 97 and ord(x) <= 122):
        consonants += 1
print("\nVowels =", vowels)
print("\nConsonants =", consonants)
```



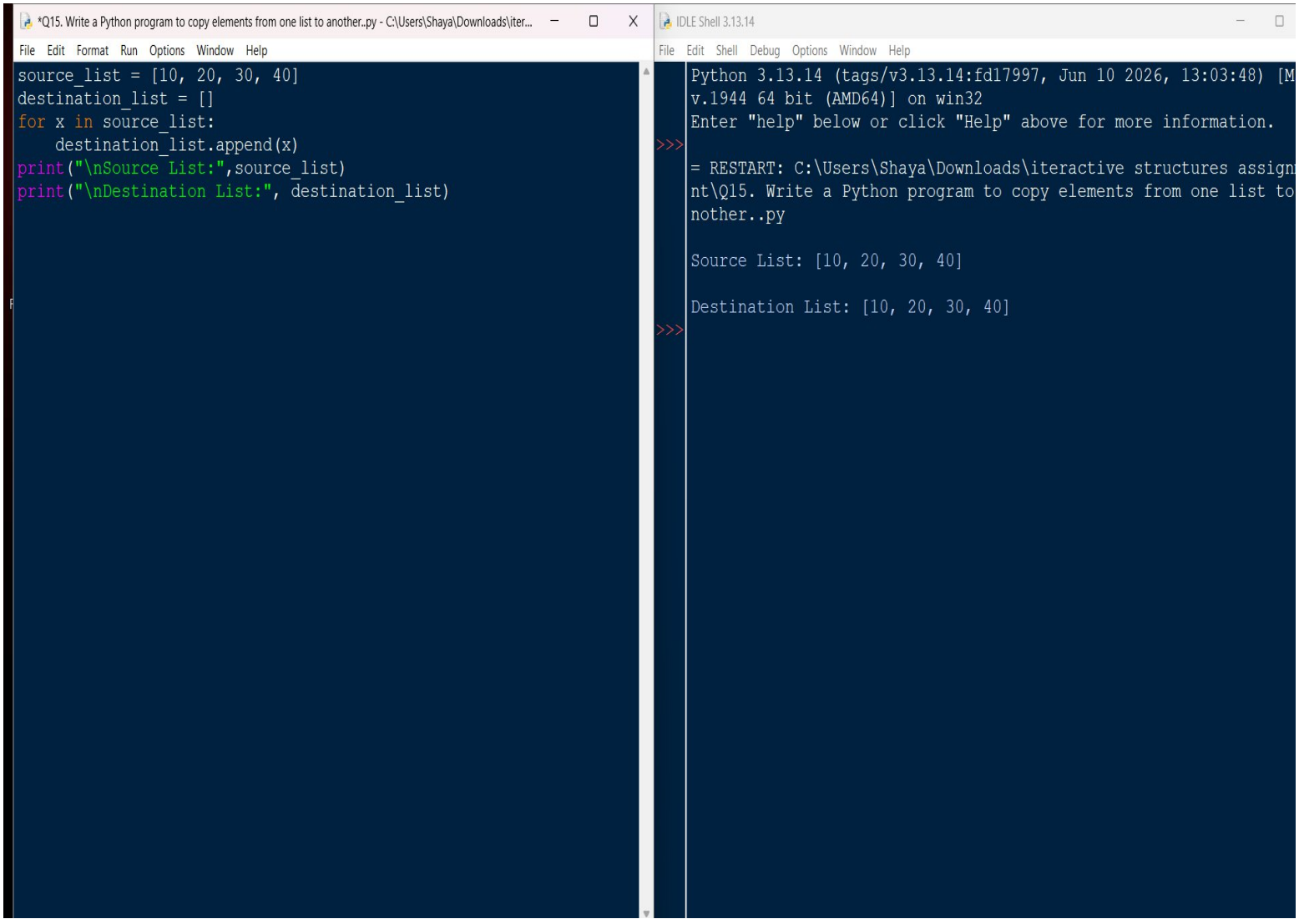
Q14. Write a Python program that generates list of values. The values must be taken from user as input.

```
numbers = []
n = int(input("\nHow many values do you want to enter? "))
for i in range(n):
    value = int(input("\nEnter value: "))
    numbers.append(value)
print("\nList =", numbers)
```



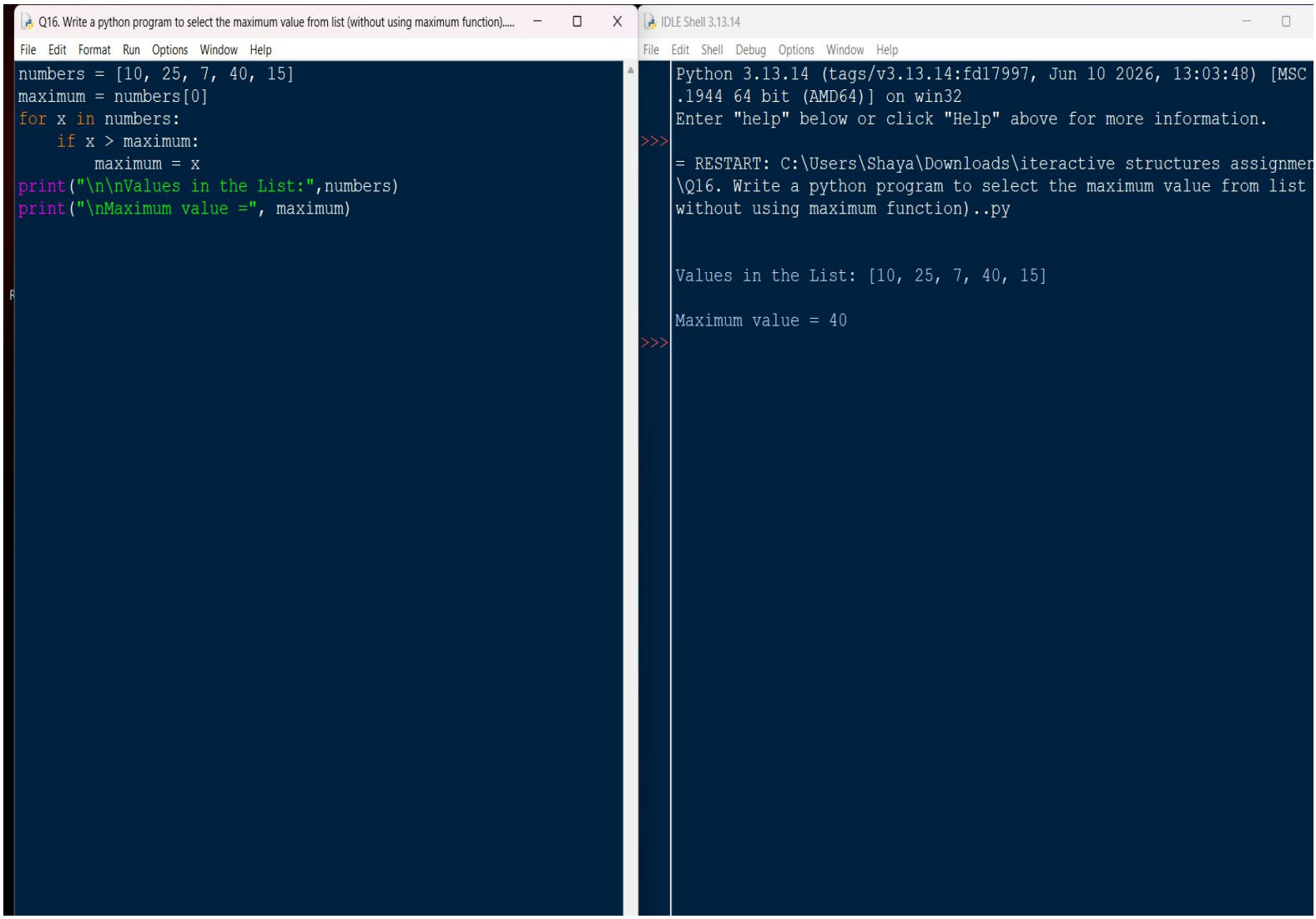
Q15 Write a Python program to copy elements from one list to another.

```
source_list = [10, 20, 30, 40]
destination_list = []
for x in source_list:
    destination_list.append(x)
print("\nSource List:",source_list)
print("\nDestination List:", destination_list)
```



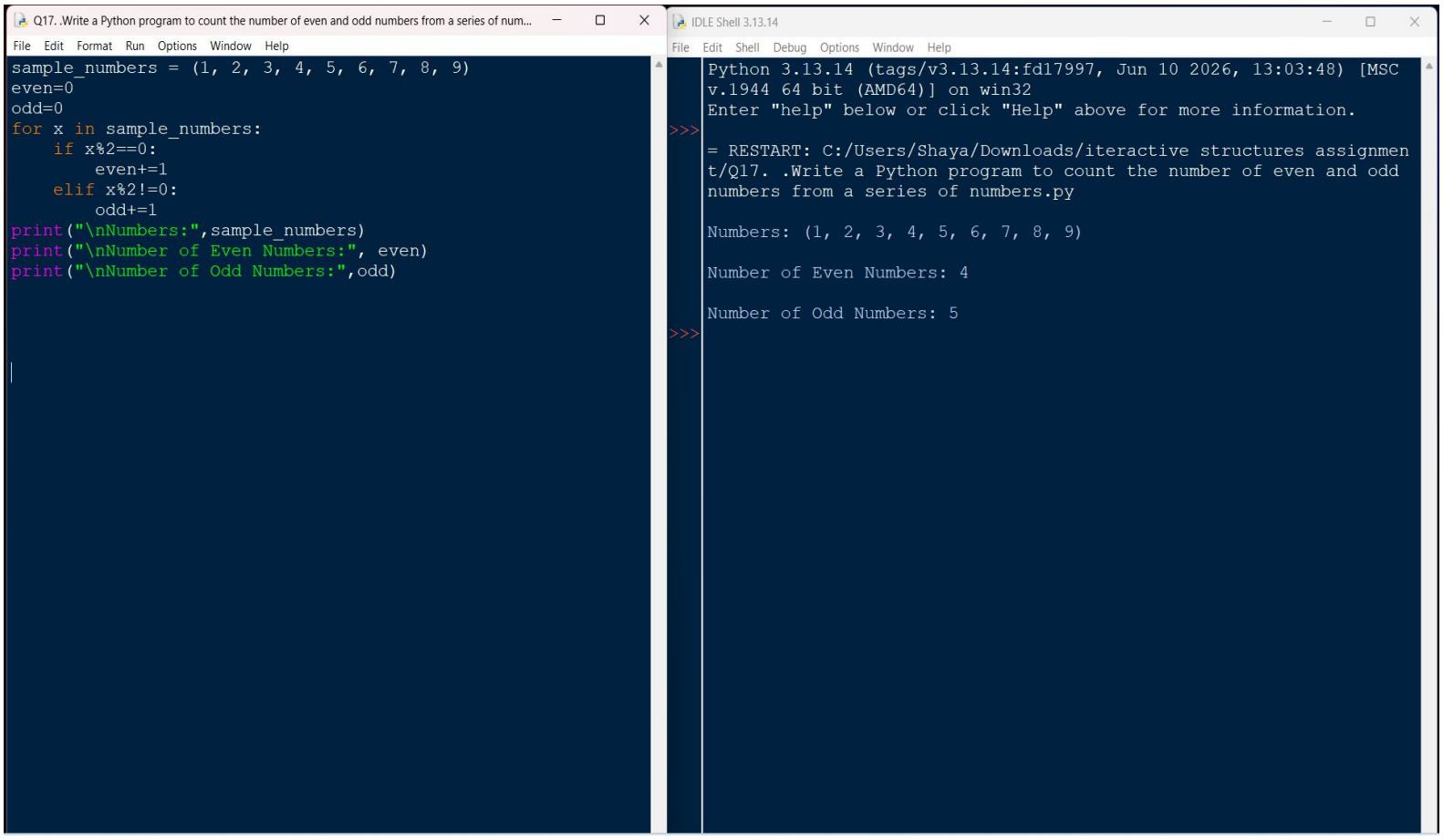
Q16. Write a python program to select the maximum value from list (without using maximum function).

```
numbers = [10, 25, 7, 40, 15]
maximum = numbers[0]
for x in numbers:
    if x > maximum:
        maximum = x
print("\n\nValues in the List:",numbers)
print("\nMaximum value =", maximum)
```



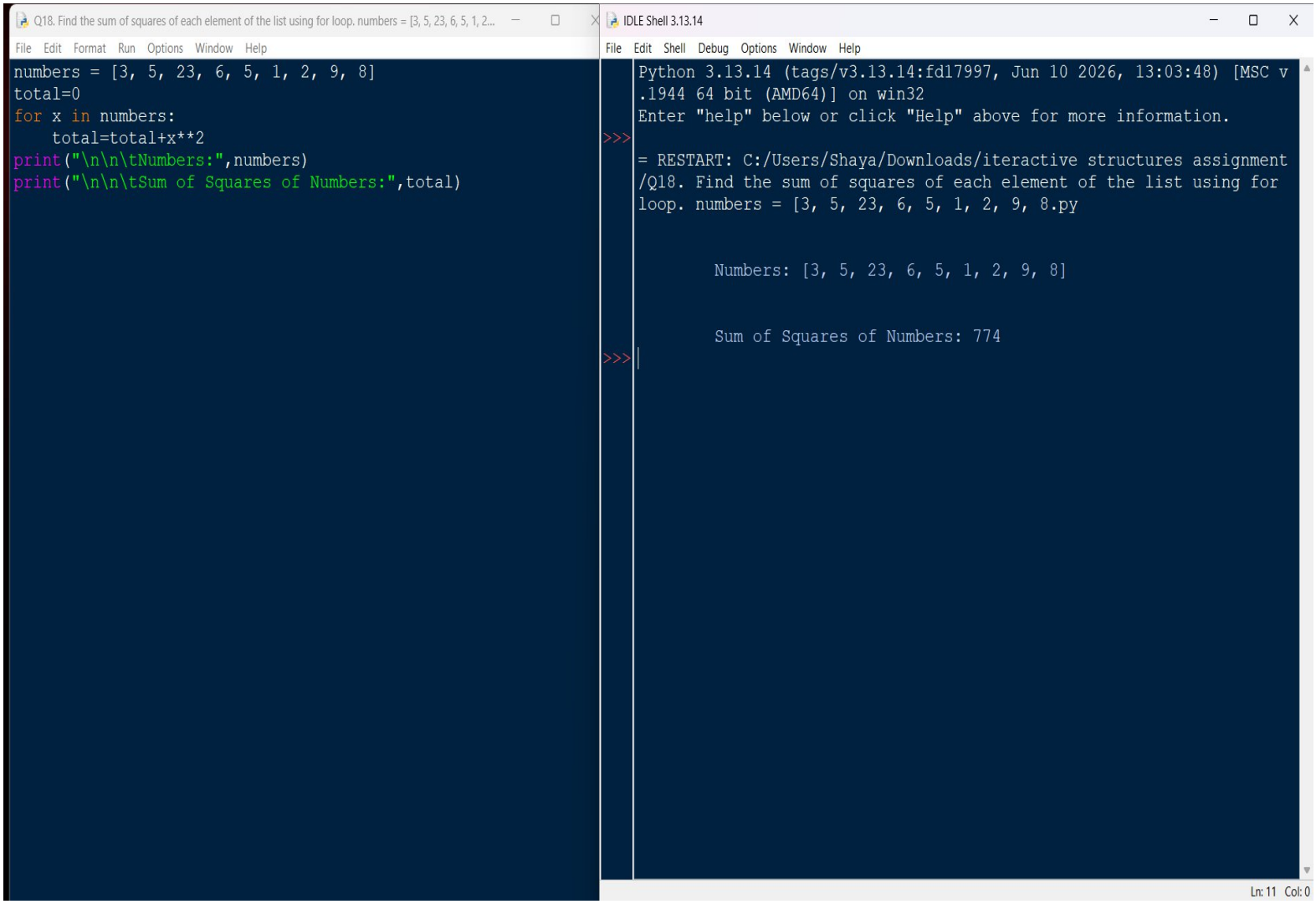
Q17. Write a Python program to count the number of even and odd numbers from a series of numbers. Sample numbers: numbers = (1, 2, 3, 4, 5, 6, 7, 8, 9)

```
sample_numbers = (1, 2, 3, 4, 5, 6, 7, 8, 9)
even=0
odd=0
for x in sample_numbers:
    if x%2==0:
        even+=1
    elif x%2!=0:
        odd+=1
print("\nNumbers:",sample_numbers)
print("\nNumber of Even Numbers:", even)
print("\nNumber of Odd Numbers:",odd)
```



Q18. Find the sum of squares of each element of the list using for loop. numbers = [3, 5, 23, 6, 5, 1, 2, 9, 8]

```
numbers = [3, 5, 23, 6, 5, 1, 2, 9, 8]
total=0
for x in numbers:
    total=total+x**2
print("\n\n\tNumbers:",numbers)
print("\n\n\tSum of Squares of Numbers:",total)
```



Q19. From given list: gadgets = [“Mobile”, “Laptop”, 100, “Camera”, 310.28, “Speakers”, 27.00, “Television”, 1000, “Laptop Case”, “Camera Lens”]
a) Create separate lists of strings and numbers.
b) Sort the strings list in ascending order
c) Sort the strings list in descending order
d) Sort the number list from lowest to highest
e) Sort the number list from highest to lowest

```
gadgets = ["Mobile", "Laptop", 100, "Camera", 310.28,
"Speakers", 27.00, "Television", 1000, "Laptop Case", "Camera
Lens"]

strings=[]

numbers=[]

for x in gadgets:

    if type(x)==str:

        strings.append(x)

    else:

        numbers.append(x)

print("\nStrings:",strings)

print("\nNumbers:",numbers)

strings.sort()

print("\nStrings in Ascending Order:",strings)

strings.sort(reverse=True)

print("\nStrings in Descending Order:",strings)

numbers.sort()

print("\nNumbers from lowest to highest:",numbers)

numbers.sort(reverse=True)

print("\nNumbers from highest to lowest:",numbers)
```

Q19. from given list in gadgets.py - C:\Users\Shaya\Downloads\iteractive structures assignment\Q19. from given list in gadgets.py (3.13.14)

File Edit Format Run Options Window Help

```
gadgets = ["Mobile", "Laptop", 100, "Camera", 310.28, "Speakers", 27.00, "Television", 1000, "Laptop Case", "Camera Lens"]
strings=[]
numbers=[]
for x in gadgets:
    if type(x)==str:
        strings.append(x)
    else:
        numbers.append(x)
print("\nStrings:",strings)
print("\nNumbers:",numbers)
strings.sort()
print("\nStrings in Ascending Order:",strings)
strings.sort(reverse=True)
print("\nStrings in Descending Order:",strings)
numbers.sort()
print("\nNumbers from lowest to highest:",numbers)
numbers.sort(reverse=True)
print("\nNumbers from highest to lowest:",numbers)
```

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\Shaya\Downloads\iteractive structures assignment\Q19. from given list in gadgets.py

Strings: ['Mobile', 'Laptop', 'Camera', 'Speakers', 'Television', 'Laptop Case', 'Camera Lens']

Numbers: [100, 310.28, 27.0, 1000]

Strings in Ascending Order: ['Camera', 'Camera Lens', 'Laptop', 'Laptop Case', 'Mobile', 'Speakers', 'Television']

Strings in Descending Order: ['Television', 'Speakers', 'Mobile', 'Laptop Case', 'Laptop', 'Camera Lens', 'Camera']

Numbers from lowest to highest: [27.0, 100, 310.28, 1000]

Numbers from highest to lowest: [1000, 310.28, 100, 27.0]
>>>
```